

Assignment-2 for the post of Web & Tech Manager, E-Cell IIT Bombay

Submission Deadline: 16th March 2026, 06:00 AM

Deadline is strict.

You are advised to complete as much of the assignment as possible. Focus on implementing the **core functionality first**, and then attempt additional improvements for bonus points.

Use of **AI tools is allowed**. However, you are expected to **fully understand every part of the code you submit**. During evaluation, you may be asked to explain your implementation.

Problem Statement

E-Cell IIT Bombay runs multiple initiatives throughout the year, including events, competitions, partnerships, marketing campaigns, media releases, and internal operations. These initiatives rely on several digital platforms such as websites, dashboards, and internal tools.

Currently, most issues and queries are reported informally through personal messages, emails, or direct communication with team members. As the organization grows, this approach becomes difficult to manage.

Some of the common challenges include:

- Queries being sent to the wrong person or department
- Lack of visibility into the progress of ongoing issues
- Duplicate reports of the same problem
- No centralized record of past queries or their resolutions
- Difficulty coordinating between managers and their teams

To address these challenges, the Web & Tech team plans to build a **centralized portal to manage queries and issues across the organization**.

The system should allow users to raise queries and allow internal teams to collaboratively resolve them.

The portal must support **multiple levels of access**, since different members of the organization should only see the queries relevant to them.

For example:

- Some users may have visibility across the entire system
- Some users may manage queries related to a specific initiative or domain
- Some users may only be able to work on queries assigned to them

Additionally, queries may involve **discussion between multiple stakeholders**, attachments such as screenshots or files, and a clear history of updates.

Your Task

Design and implement a **Query and Issue Management Portal** that addresses the above problem.

Your solution should demonstrate:

- strong **system design thinking**
- good **engineering practices**
- attention to **product quality**

You are free to choose appropriate technologies, architectures, and design patterns.

Part 1 - System Design

Explain how you would design the system.

Your explanation may include (but is not limited to):

- how queries should be represented and organized
- how different users interact with the system
- how access control should work
- how queries should be assigned or routed
- how conversations or updates within a query should be handled
- how attachments such as screenshots or files should be stored

Clearly justify your design choices.

Part 2 - Access Control

The portal must support **multiple levels of access and responsibility**.

Design a permission system that enables:

- hierarchical access
- controlled visibility of queries
- delegation of responsibilities between users

Explain how this model would function in practice and how unauthorized access will be prevented.

Part 3 - Implementation

Build a **working prototype** of the portal.

Your implementation should include:

- a user interface for creating and managing queries
- a backend system to manage users and queries
- authentication and authorization
- the ability to attach screenshots or files
- a mechanism for updating or discussing queries

Allowed Technology Stacks

Frontend

- Angular
- React

Backend

- Django
- Node.js (Express or NestJS)

Database

- Any database of your choice (Postgres/mysql/MongoDB)

Product Quality Requirements

Your implementation must meet the following standards:

- The platform should look **professional and production-ready**
- The UI should **not resemble generic AI-generated interfaces**
- The interface must be **fully responsive across devices and screen sizes**
- Proper **error handling** must be implemented on both frontend and backend
- The application should be **stable and free from obvious bugs**

Attention to **UI/UX quality, usability, and robustness** will be considered during evaluation.

Engineering Understanding Requirement

You are expected to **fully understand every part of the system you implement**.

During evaluation, you may be asked to explain:

- how different parts of your system work
- why you chose specific design decisions
- how your code handles edge cases or failures
- how different components interact with each other

You should **not rely on code that you cannot explain or justify**.

Part 4 - Additional Feature

Propose and implement **at least one feature** that significantly improves the usefulness of the portal.

This feature should demonstrate **thoughtful engineering** and add meaningful value to the system.

Explain:

- what the feature does
 - why it improves the system
 - how it is implemented
-

Submission Guidelines

1. Create a **GitHub repository** with the following naming format:

<Name>_Web_Assignment2026

Add the following users as collaborators: masterking008, Arnavbhau56

All code related to the assignment should be pushed to this repository.

2. Prepare a **document containing answers to all the questions** (including the textual explanations from all parts of the assignment).

Export this document as a **PDF**.

3. Create a **ZIP file** containing:
 - All the code from the repository
 - The final PDF document

The **ZIP file and PDF name must follow this format**:

<RollNo>_<Name>_Web

4. Email the ZIP file and the GitHub repository link to:

dineshsahu.ecell@gmail.com

CC: 45arnav@gmail.com, priyanshugehlot011@gmail.com, vedpatil.ecell.iitb@gmail.com

Important Notes

- The **submission deadline is strict**. Late submissions may not be considered.
- Focus on building **clean, well-structured, and maintainable code**.
- Implement the **core functionality first**, and then attempt improvements.
- Use of **AI tools is allowed**, but you must be able to explain your implementation.

Contact for Queries

If you have any questions regarding the assignment or submission process, feel free to contact:

Dinesh

+91 8109286424

Arnav

+91 9414454858